

MLB Hitters Hub — Full Feature Implementation

Agent build instructions for Antigravity IDE | Sports betting analytics platform module

Copy/paste note: This entire document is a single instruction set for an AI coding agent (e.g. Antigravity IDE Agent). It can be pasted in full, or this PDF can be attached/uploaded directly as the task brief.

You are a senior full-stack engineer.

Build a production-ready MLB Hitters Hub module inside an existing sports betting analytics application.

Existing Stack

Frontend

- React
- Next.js
- Tailwind CSS
- Recharts
- Hosted on Vercel

Backend

- FastAPI
- Python
- Pandas
- JSON-based data processing

The system already receives fresh CSV files daily via GitHub Actions.

Your implementation should focus on:

- Python data processing
 - Pandas dataframe logic
 - JSON APIs
 - FastAPI endpoints
 - React/Next.js frontend components
-

Data Sources

The backend receives these CSV files daily:

hit_pred.csv

Columns:

Batter
Pitcher
Game
HRF
Pred
Trend

O/U
Odds
AB
H
HR
BA
OBP
OPS

Example:

N. Kurtz, 1B (L)
W. Urena (R)
ANA @ ATH
77%
1, 3, 1, 1, 1
0.5

hr_pred.csv

Columns:

Batter
Pitcher
Game
HRF
HR Pred
Trend
Odds
AB
H
HR
BA
OBP
OPS

tb_pred.csv

Columns:

Batter
Pitcher
Game
HRF
TB Pred
Trend
Line
Over
AB
H
HR
BA
OBP
OPS

bvp.csv

Columns:

Batter
Pitcher
Game
HRF
qAB
HH%
AB
H
2B/3B
HR
BB
BA
OBP
OPS

last7hitting.csv

Columns:

Name
Team
G
PA
AB
R
H
HR
RBI
AVG
OBP
SLG
OPS

Backend Requirements

Create FastAPI endpoints.

Endpoint 1

/api/hitters/hits

Returns hit projections.

Endpoint 2

/api/hitters/hr

Returns HR projections.

Endpoint 3

/api/hitters/tb

Returns total base projections.

Endpoint 4

`/api/hitters/bvp`

Returns batter vs pitcher data.

Endpoint 5

`/api/hitters/last7`

Returns last 7 day hitters.

Game Selector

Generate all unique games from:

Game

Examples:

ANA @ ATH
MIL @ ATL
TOR @ CHN

Create endpoint:

`/api/hitters/games`

Returns unique games.

Frontend dropdown should allow selecting:

ALL Games
ANA @ ATH
MIL @ ATL
TOR @ CHN

Selecting a game filters every table.

Team Logos

Parse game strings.

Example:

ANA @ ATH

Extract:

ANA
ATH

Create team logo mapping.

Display logos in game selector.

100% Club

Create dedicated section.

Add buttons:

Hits
Total Bases

No HRR support. Do NOT implement HRR.

Hits Club Logic

Use:

Trend

Example:

1, 3, 1, 1, 1

Use:

0/U

Example:

0.5

Player qualifies if all five games beat the prop line.

Example:

[1, 3, 1, 1, 1]
line = 0.5

Result:

5/5

Player appears in club.

Total Bases Club Logic

Use:

Trend

and

Line

Example:

trend = [4, 3, 5, 5, 2]
line = 1.5

If all five games exceed the line:

5/5

Player appears.

5-Day Hit Streak Section

Create:

5-Day Hit Streak Club

Use only hit_pred.csv.

Logic:

```
trend = [1,3,1,1,1]
```

If all five games contain at least one hit:

```
all(x > 0)
```

Player qualifies.

Important: Do NOT attempt true 5+ streak calculations. Only use available five-game trend data. Label section **5-Day Hit Streak Club**, not **5+ Day Streak**.

Bet Sheet

Every player row should include:

+

green action button.

Clicking adds player to bet sheet.

Bet Sheet Requirements

- Store selections in browser localStorage.
- Show side panel.

Example:

Bet Sheet

Aaron Judge

Shohei Ohtani

Freddie Freeman

Allow:

- Remove selection
- Clear all

Midnight Reset

Bet sheet automatically resets daily.

Implementation:

```
localStorage:
{
  date: "YYYY-MM-DD",
  selections: [...]
}
```

When current date differs:

clear selections

No cron jobs. No backend jobs. Client-side only.

Frontend UI

Create a dark sports analytics dashboard.

Use:

- Tailwind CSS
- Responsive layout
- Mobile friendly
- Desktop friendly

Sections:

- Game Selector
- 100% Club
- 5-Day Hit Streak Club
- Hits Projections
- HR Projections
- Total Bases Projections
- Batter vs Pitcher
- Last 7 Day Hot Hitters
- Bet Sheet

Data Processing

Create reusable pandas services.

Requirements:

- CSV loading service
- Data normalization
- JSON serialization
- Filtering by game
- Trend parsing
- 100% club calculations
- Streak calculations

Avoid duplicated logic. Use clean architecture.

Deliverables

Generate:

Backend

- FastAPI routes
- Services
- Models
- Utilities

Frontend

- Next.js pages
- Components
- Hooks
- API integrations
- Tailwind styling

Documentation

Create a comprehensive README.md containing:

- Project structure
- Environment variables
- Local development setup
- FastAPI startup instructions
- Next.js startup instructions
- CSV data directory setup
- Vercel deployment guide
- Build commands
- Production deployment steps
- Troubleshooting section

Include all code necessary for deployment and execution.

The final result should be production-ready and deployable to Vercel with a FastAPI backend.